# Variance and Concentration

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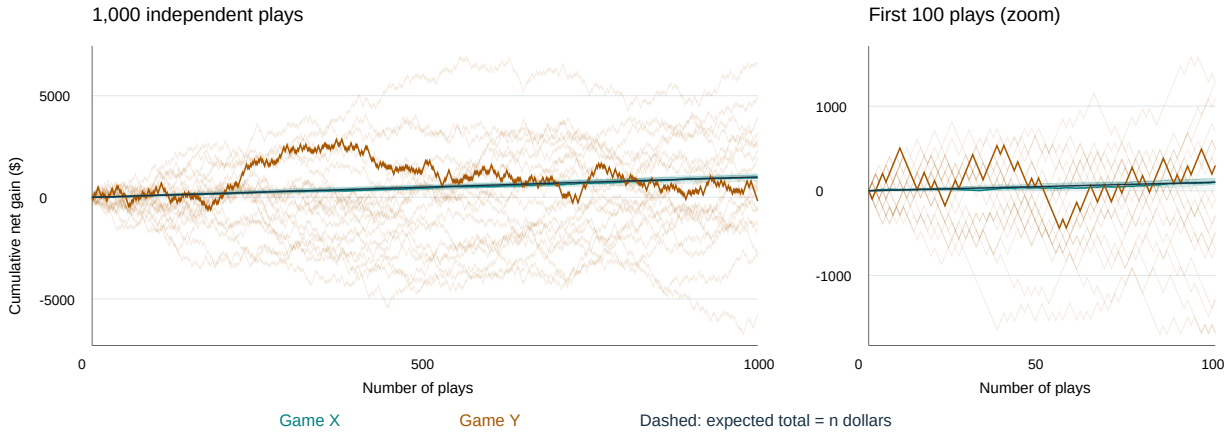
COMP2711 Discrete Mathematical Tools

## 1 Variance

Two games give the following net gains in dollars:

| | possible gains | probabilities |
|---|---|---|
| $X$ | $-2,\ 4$ | $1/2,\ 1/2$ |
| $Y$ | $-98,\ 100$ | $1/2,\ 1/2$ |

Both have expectation 1: $\mathbb{E}[X] = (-2+4)/2 = 1$ and $\mathbb{E}[Y] = (-98+100)/2 = 1$. Game $Y$ has much larger gains and losses, despite having the same mean. Expectation describes the average value; it does not describe the spread. All examples are discrete. Whenever a variance is used, assume it is finite.




Cumulative gains from independent plays. The dashed line is the expected total gain; the right panel enlarges the first 100 plays.

## 1.1 Definition and calculation

The average signed deviation cannot measure spread: $\mathbb{E}[X - \mathbb{E}[X]] = 0$. Positive and negative deviations cancel. Squaring the deviations avoids this cancellation.

For $\mu = \mathbb{E}[X]$, the **variance** is

$$\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \sum_x (x-\mu)^2 \Pr[X = x].$$

Variance is the probability-weighted average of squared distances from the mean. It is nonnegative. It is zero exactly when $X = \mu$ with probability one. For the two games,

$$\mathrm{Var}(X) = \tfrac{1}{2}(-2-1)^2 + \tfrac{1}{2}(4-1)^2 = 9, \qquad \mathrm{Var}(Y) = \tfrac{1}{2}(-98-1)^2 + \tfrac{1}{2}(100-1)^2 = 9801.$$

The **standard deviation** is $\sigma = \sqrt{\mathrm{Var}(X)}$. Variance has squared units; standard deviation has the same units as $X$. Thus the games have standard deviations \$3 and \$99.

**A formula for computing variance.**

By linearity of expectation and the fact that $\mu$ is constant,

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2.\end{aligned}$$

For game $Y$, $\mathbb{E}[Y^2] = \frac{1}{2}(-98)^2 + \frac{1}{2}(100)^2 = 9802$, so $\mathrm{Var}(Y) = 9802 - 1^2 = 9801$.

**A fair die.**

Let $D$ be the number shown on a fair six-sided die. Then

$$\mathbb{E}[D] = \frac{1+2+3+4+5+6}{6} = \frac{7}{2}, \qquad \mathbb{E}[D^2] = \frac{1+4+9+16+25+36}{6} = \frac{91}{6}.$$

Therefore $\mathrm{Var}(D) = 91/6 - (7/2)^2 = 35/12$.

## 1.2 Shifting and scaling

For real constants $a, b$,

$$\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X).$$

**Proof (exercise from the slides).** Write $\mu = \mathbb{E}[X]$. By linearity, $\mathbb{E}[aX + b] = a\mu + b$. Therefore

$$\begin{aligned}\mathrm{Var}(aX+b) &= \mathbb{E}[(aX + b - (a\mu + b))^2] \\ &= \mathbb{E}[a^2(X-\mu)^2] = a^2 \mathrm{Var}(X).\end{aligned}$$

Taking $a = 1$ gives $\mathrm{Var}(X + b) = \mathrm{Var}(X)$; taking $b = 0$ gives $\mathrm{Var}(aX) = a^2 \mathrm{Var}(X)$. Adding a constant shifts every value and its mean together, so spread is unchanged. Multiplying by $a$ multiplies squared distances by $a^2$. For game $X$, $\mathrm{Var}(X + 10) = 9$ and $\mathrm{Var}(2X) = 36$.

## 1.3 Independent sums

Recall that discrete $X, Y$ are independent if, for all possible $x, y$,

$$\Pr[X = x \text{ and } Y = y] = \Pr[X = x] \Pr[Y = y].$$

**Proof of $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ (exercise from the slides).** Using the joint distribution and independence,

$$\begin{aligned}
\mathbb{E}[XY] &= \sum_{x,y} xy \Pr[X = x,\ Y = y] \\
&= \sum_{x,y} xy \Pr[X = x] \Pr[Y = y] \\
&= \left( \sum_{x} x \Pr[X = x] \right) \left( \sum_{y} y \Pr[Y = y] \right) = \mathbb{E}[X]\mathbb{E}[Y].
\end{aligned}$$

For countably infinite supports, finite second moments ensure absolute convergence: $\mathbb{E}[|XY|] \leq (\mathbb{E}[X^2] + \mathbb{E}[Y^2])/2 < \infty$. Consequently,

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y).$$

To see where independence is used, first expand without assuming it:

$$\begin{aligned}
\operatorname{Var}(X + Y) &= \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2] - (\mathbb{E}[X] + \mathbb{E}[Y])^2 \\
&= \operatorname{Var}(X) + \operatorname{Var}(Y) + 2(\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]).
\end{aligned}$$

The quantity $\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$ is the covariance, $\operatorname{Cov}(X, Y)$. For independent variables it is zero. For a finite family, pairwise independence is sufficient for variance addition. Independence is a sufficient condition, not a necessary one.

**Independent trials versus copying one result.**

Let $I, J$ be the head indicators of two independent fair flips. Each has mean $1/2$ and variance $\frac{1}{2}(0 - \frac{1}{2})^2 + \frac{1}{2}(1 - \frac{1}{2})^2 = 1/4$. Then $\operatorname{Var}(I + J) = 1/2$. But copying the same result gives $I + I = 2I$, whose variance is 1. Linearity of expectation does not need independence; this variance-addition rule does. For independent dice, $\operatorname{Var}(D_1 + D_2) = 35/6$.

**Binomial variance.**

For a trial that says Yes with probability $p$, define $I = 1$ for Yes and $I = 0$ for No. Because $I^2 = I$,

$$\mathbb{E}[I^2] = \mathbb{E}[I] = p, \qquad \operatorname{Var}(I) = p - p^2 = p(1 - p).$$

If $H \sim \operatorname{Bin}(n, p)$, write $H = I_1 + \cdots + I_n$ for the independent trial indicators. Then

$$\mathbb{E}[H] = np, \qquad \operatorname{Var}(H) = np(1 - p).$$

For 100 independent fair flips, the head count has mean 50 and variance 25, so its standard deviation is 5. The formula also gives variance $3/4$ for the number of heads in three independent fair flips.

# 2 Concentration

Concentration bounds limit the probability of a large deviation. They are useful when computing the full distribution is difficult.

**Example: Items that stay in place.**

Uniformly permute $n \geq 10$ items, and let $X$ count items that stay in their original positions. If $I_i$ indicates that item $i$ stays in place, then

$$X = \sum_{i=1}^{n} I_i, \qquad \mathbb{E}[X] = \sum_{i=1}^{n} \Pr[I_i = 1] = n \cdot \frac{1}{n} = 1.$$

To calculate $\Pr[X > 10]$ exactly, we would need to count permutations with each possible number of fixed points. Exactly $j$ fixed points can be chosen in $\binom{n}{j}$ ways; the remaining items must form a derangement. Markov's inequality gives a bound using only the mean.

## 2.1 Markov's inequality

**Theorem.** If $X$ is nonnegative and $0 < \mu = \mathbb{E}[X] < \infty$, then for every $k \geq 1$,

$$\Pr[X \geq k\mu] \leq \frac{1}{k}.$$

**Proof.** Suppose, for a contradiction, that $\Pr[X \geq k\mu] > 1/k$. On this event, each value is at least $k\mu$; all other contributions to expectation are nonnegative. Thus

$$\mathbb{E}[X] \geq k\mu \Pr[X \geq k\mu] > k\mu \cdot \frac{1}{k} = \mu,$$

contradicting $\mathbb{E}[X] = \mu$.

The same argument gives the threshold form: for nonnegative $X$ with finite expectation and any $c > 0$,

$$\Pr[X \geq c] \leq \frac{\mathbb{E}[X]}{c}.$$

Nonnegativity is essential because negative contributions could otherwise cancel positive ones.

**Applying Markov to the permutation.**

Since $X$ is integer-valued, $\{X > 10\} = \{X \geq 11\}$. Therefore

$$\Pr[X > 10] \leq \frac{\mathbb{E}[X]}{11} = \frac{1}{11} < \frac{1}{10}.$$

This proves the strict inequality stated in the slides.

## 2.2 Chebyshev's inequality

Apply Markov to the squared distance from the mean, $Z = (X - \mu)^2$. This is nonnegative and has expectation $\mathrm{Var}(X)$.

**Theorem.** Let $X$ have mean $\mu$ and variance $0 < \sigma^2 < \infty$. For every $t \geq 1$,

$$\Pr[|X - \mu| \geq t\sigma] \leq \frac{1}{t^2}.$$

**Proof.**

$$\begin{aligned}\Pr[|X - \mu| \geq t\sigma] &= \Pr[(X - \mu)^2 \geq t^2\sigma^2] \\ &\leq \frac{\mathbb{E}[(X - \mu)^2]}{t^2\sigma^2} = \frac{1}{t^2}.\end{aligned}$$

Equivalently, for any $\varepsilon > 0$,

$$\Pr[|X - \mu| \geq \varepsilon] \leq \frac{\sigma^2}{\varepsilon^2}.$$

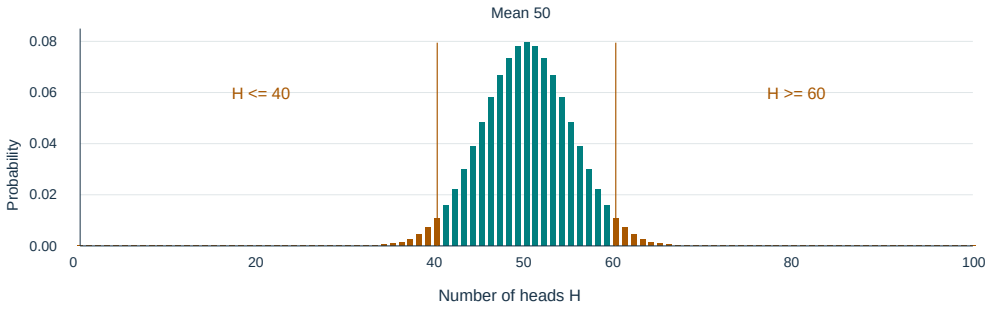
This threshold form also holds when $\sigma^2 = 0$, since then $X = \mu$ with probability one.

Markov uses the mean (a first-moment method); Chebyshev uses the variance (a second-moment method). Chebyshev does not require $X \geq 0$, because it applies Markov to the squared distance.

**Example: 100 fair flips.**

For $H \sim \mathrm{Bin}(100, 1/2)$, the mean is 50, variance 25, and standard deviation 5. Thus

$$\Pr[H \leq 40 \text{ or } H \geq 60] = \Pr[|H - 50| \geq 2 \cdot 5] \leq \frac{1}{4}.$$




The binomial distribution, with the two tail events highlighted.

# 3 Independent averages and the law of large numbers

## 3.1 The variance of an average

Let $X_1, \ldots, X_n$ be independent, each with mean $\mu$ and finite variance $\sigma^2$. Define

$$\overline{X}_n = \frac{X_1 + \cdots + X_n}{n}.$$

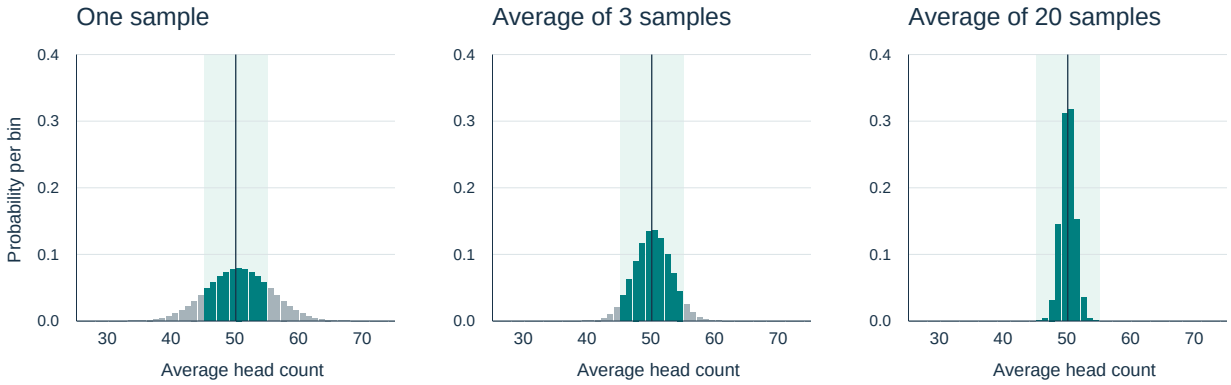
Then

$$\mathbb{E}[\overline{X}_n] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i] = \mu, \qquad \mathrm{Var}(\overline{X}_n) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{\sigma^2}{n}.$$

Its standard deviation is $\sigma/\sqrt{n}$. The variance of a *sum* grows with $n$, while the variance of the *average* shrinks with $n$. Four times as many independent observations halve the standard deviation. Copying the same observation does not reduce variance.

**Example: Averaging binomial observations.**

If each $X_i \sim \mathrm{Bin}(100, 1/2)$, every sample average has mean 50. Averaging 3 observations gives variance $25/3$; averaging 20 gives variance $25/20$.




These are distributions of the sample average, not individual simulated runs.

## 3.2 Law of large numbers

Let $X_1, X_2, \ldots$ be independent samples from the same distribution, with finite mean $\mu$ and variance $\sigma^2$. For every fixed $\varepsilon > 0$, Chebyshev applied to the average gives

$$\Pr[|\overline{X}_n - \mu| \geq \varepsilon] \leq \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0.$$

This is the **weak law of large numbers**, in its finite-variance form. The probability of a fixed-size error tends to zero as the number of independent samples grows. It justifies the previous lecture's interpretation of expectation as the long-run average.

It does not say the average improves after every new observation, or that the next result compensates for past results. After ten heads on independent fair flips, the next flip is still heads with probability $1/2$.

# 4 Exercises

Exercises 1–5 are the practice questions from the slides; Exercises 6–8 provide further practice. The proofs of shifting and scaling and of $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ under independence are included above.

1. A fair coin is flipped twice independently. Let $X$ count heads. Find $\mathbb{E}[X^2]$, $\mathrm{Var}(X)$, and $\mathrm{Var}(3X - 2)$.
2. Independent $X, Y$ satisfy $\mathbb{E}[X] = 3$, $\mathbb{E}[Y] = 5$, $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 9$. Find the mean and variance of $2X - Y + 1$. Which calculation remains valid without independence?
3. A nonnegative queue length $Q$ has expectation 2. Give a Markov bound on $\Pr[Q \geq 8]$. Find a distribution attaining the bound.
4. A coin has unknown head probability $p$. How many independent flips suffice, using Chebyshev, to make the probability of error at least 0.01 in the head fraction at most 0.05?
5. Let $I$ be a fair-coin head indicator and set $X_i = I$ for every $i$. Find the mean and variance of $\overline{X}_n$. For $\varepsilon = 1/4$, find $\Pr[|\overline{X}_n - 1/2| \geq \varepsilon]$. Why does the averaging argument not apply?
6. A random variable takes values $-1, 0, 2$ with probabilities $1/4, 1/2, 1/4$. Find its mean, variance, and standard deviation.
7. Let $H \sim \mathrm{Bin}(100, 1/2)$. Bound $\Pr[H \geq 70]$ using Markov, then using Chebyshev. Which bound is smaller?
8. Independent observations each have mean $\mu$ and variance 9. How many observations suffice to guarantee $\Pr[|\overline{X}_n - \mu| < 0.5] \geq 0.99$ using Chebyshev?

## 5 Solutions

1. The probabilities of $0, 1, 2$ are $1/4, 1/2, 1/4$, so

$$\mathbb{E}[X] = 1, \quad \mathbb{E}[X^2] = 0 + \tfrac{1}{2} + 4(\tfrac{1}{4}) = \tfrac{3}{2}, \quad \mathrm{Var}(X) = \tfrac{3}{2} - 1 = \tfrac{1}{2}.$$

   Then $\mathrm{Var}(3X - 2) = 3^2 \mathrm{Var}(X) = 9/2$.

2. Linearity gives $\mathbb{E}[2X - Y + 1] = 2(3) - 5 + 1 = 2$. Independence gives $\mathrm{Var}(2X - Y + 1) = 2^2(4) + (-1)^2(9) = 25$. The expectation calculation remains valid without independence; the variance is no longer determined by the given data alone.

3. Markov gives $\Pr[Q \geq 8] \leq 2/8 = 1/4$. Let $Q = 8$ with probability $1/4$ and $Q = 0$ otherwise. Then $\mathbb{E}[Q] = 8/4 = 2$ and $\Pr[Q \geq 8] = 1/4$, attaining the bound.

4. Let $H \sim \mathrm{Bin}(n, p)$ count heads and set $\widehat{p} = H/n$. Then $\mathbb{E}[\widehat{p}] = p$ and

$$\mathrm{Var}(\widehat{p}) = \frac{p(1-p)}{n} \leq \frac{1}{4n},$$

   because $p(1-p) = 1/4 - (p - 1/2)^2 \leq 1/4$. Chebyshev gives

$$\Pr[|\widehat{p} - p| \geq 0.01] \leq \frac{1}{4n(0.01)^2}.$$

   Requiring this to be at most 0.05 gives $n \geq 1/(4(0.05)(0.01)^2) = 50{,}000$. Taking complements gives a probability of at least 0.95 that the error is below 0.01.

5. $\overline{X}_n = (I + \cdots + I)/n = I$. Its mean is $1/2$ and variance $1/4$. Its distance from $1/2$ is always $1/2$, so the requested probability is 1. These are copies of the same outcome, not independent observations.

6. The mean is $\mu = -1/4 + 2/4 = 1/4$, and $\mathbb{E}[X^2] = 1/4 + 4/4 = 5/4$. Therefore

$$\mathrm{Var}(X) = \frac{5}{4} - \left(\frac{1}{4}\right)^2 = \frac{19}{16}, \qquad \sigma = \frac{\sqrt{19}}{4}.$$

7. Here $\mathbb{E}[H] = 50$ and $\mathrm{Var}(H) = 25$. Markov gives $\Pr[H \geq 70] \leq 50/70 = 5/7$. Since $\{H \geq 70\} \subseteq \{|H - 50| \geq 20\}$, Chebyshev gives

$$\Pr[H \geq 70] \leq \Pr[|H - 50| \geq 20] \leq \frac{25}{20^2} = \frac{1}{16}.$$

   Chebyshev gives the smaller bound.

8. The sample average has variance $9/n$. Thus

$$\Pr[|\overline{X}_n - \mu| \geq 0.5] \leq \frac{9}{n(0.5)^2} = \frac{36}{n}.$$

   Requiring $36/n \leq 0.01$ gives $n \geq 3600$. Taking complements proves the guarantee.